

Bradycardia (with pulses)

History

- Does the patient have bradycardia? (HR < 50)
- Is the patient symptomatic?
- If so, when did the symptoms start?
- What is the patient's medical history? – Cardiac or kidney disease? Diabetes? Hypothyroidism? Pacemaker?
- Has the patient had head, neck or back trauma?
- Has the patient had a toxic exposure or overdose?
- Does the patient have altered mental status or stroke symptoms?
- What medications does the patient take? (see list)

Signs and Symptoms of Hemodynamically Significant Bradycardia (Shock)

- Hypotension (SBP <90)
- Syncope
- Dizziness / lightheadedness (near-syncope)
- Altered mental status / agitation
- Chest pain
- Shortness of breath / difficulty breathing
- Signs of pulmonary edema – crackles / rales, hypoxia
- Diaphoresis
- Cyanosis
- Decreased capillary refill

Medications That Can Cause Bradycardia

- **Beta-blockers** (metoprolol, atenolol, labetalol, sotalol, propranolol, carvedilol, pindolol, nadolol)
- **Calcium channel blockers** (amlodipine, diltiazem, felodipine, isradipine, nicardipine, nifedipine, verapamil)
- **Digitalis** (Digoxin, Lanoxin, Digitek, Lanoxicaps) **GO TO Hyperkalemia-Digoxin Toxicity Protocol**

EMT/AEMT

- Follow General Medical Care Guideline
- Follow Adult Resuscitation Guideline, as indicated

EMT/AEMT - Expanded Scope

- **NS/LR** 500ml IO if unable to tolerate fluid by mouth

Paramedic

Signs / Symptoms of Shock

- Manage airway as indicated
- 12-Lead EKG
- **NS** 500-1000 mL IV bolus if SBP ≤ 90
- **Atropine** 0.5mg – 1 mg IO/IV – repeat to max dose of 3 mg
 - If 2° Type II, 3° heart block consider immediate evacuation for immediate pacing
 - ***If Hyperkalemia suspected see below***
- **Epinephrine Push Dose** (1:100,000) 10-20 mcg (1-2 mL) IO/IV every 2-3 minutes PRN (max total 100 mcg / 10 mL) – Consult Medical Director for further doses
- Treat STEMI per appropriate guideline

***If Patient has history of renal failure, active cancer radiation or chemotherapy, or concern for rhabdomyolysis or hyperkalemia – consider aggressive early treatment as follows: ***

- **Calcium Chloride** 1g IO/IV (Do not give if taking digoxin)
- **Albuterol** 7.5 mg/9 ml nebulized or instilled down ET tube